Comprehensive Design Study of a 2026-Compliant Formula 1 Car

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1 Introduction

The evolution of propulsion systems in Formula 1 represents a continuous pursuit of peak thermodynamic efficiency intertwined with the demands of highly competitive motorsport. Historically, the sport relied entirely on high-revving, naturally aspirated internal combustion engines (ICE) that prioritized peak power output over fuel efficiency. However, a sweeping paradigm shift occurred in 2014 with the introduction of the V6 turbo-hybrid era. This regulation change mandated the integration of complex Energy Recovery Systems (ERS), comprising the Motor Generator Unit-Kinetic (MGU-K) and the Motor Generator Unit-Heat (MGU-H), operating in tandem with the internal combustion engine. While this marked a monumental step toward road-relevant hybridization and record-breaking thermal efficiency, the combustion engine remained the dominant source of propulsive power, with the electrical components serving primarily as supplementary performance enhancers.

The 2026 regulatory cycle introduces a fundamental turning point in the sport's technical trajectory, decisively redefining the balance of power generation. These new regulations mandate a drastic shift toward electrical propulsion, establishing a near-equal power split between the internal combustion engine and the electrical system. The elimination of the complex MGU-H is counterbalanced by a substantial upgrade to the MGU-K, which will see its deployment capacity nearly triple to 350 kilowatts (kW). Concurrently, the energy storage system is scaled to manage an operational capacity of approximately 4 megajoules (MJ) per lap. This radical architectural restructuring dictates that the electrical powertrain transition from a supporting role to a primary driver of performance, while the ICE transitions to utilize fully sustainable drop-in fuels to further the championship's net-zero carbon objectives.

This unprecedented reliance on electrical energy introduces a myriad of profound engineering challenges. The packaging of the heavily uprated MGU-K, alongside the significantly larger and denser battery pack within the chassis, requires meticulous spatial optimization. This electrical mass significantly influences the vehicle's weight distribution, moment of inertia, and center of gravity. Furthermore, thermal management becomes an acute constraint; rapidly dissipating the immense heat flux generated by a 350 kW electrical motor and a high-cycling 4 MJ battery mandates aggressive cooling architectures that inherently conflict with aerodynamic drag reduction goals. Energy management strategies must also be entirely overhauled. The increased electrical deployment ratio necessitates highly sophisticated, algorithmic deployment and

harvesting profiles to prevent severe power derating—and subsequent velocity drop-offs—at the end of long straights. Aerodynamically, the 2026 regulations mandate the implementation of active aerodynamic systems to mitigate drag and compensate for the altered power delivery characteristics, adding critical complexity to the vehicle's dynamic behavior.

The purpose of this paper is to present a comprehensive, multi-disciplinary design study of a 2026-compliant Formula 1 car. By analyzing the symbiotic relationship between the drastically altered power unit architecture and the new generative aerodynamic regulations, this study systematically explores optimal solutions for packaging, thermal management, and energy deployment. Utilizing advanced computational fluid dynamics (CFD) and transient lap time simulation (LTS), we evaluate the cascading effects of the 350 kW electrical architecture on chassis dynamics and overall vehicle concept. Ultimately, this research aims to establish a theoretical engineering framework for maximizing competitive performance under the constraints of the most electrically dominant regulatory cycle in the history of the sport.

2 2026 FIA Technical Regulations Overview

The 2026 Formula 1 technical regulations represent one of the most comprehensive and radical overhauls in the championship's history, simultaneously addressing power unit architecture, chassis design, and aerodynamic philosophy. Dictated by the Federation Internationale de l'Automobile (FIA) [**?**], these mandates aim to drastically reduce the carbon footprint of the sport, increase the road relevance of the technology, and enhance the competitive spectacle by enabling cars to follow each other more closely. This section delineates the core regulatory changes that govern the vehicle design space for the 2026 technical cycle.

2.1 Power Unit Regulations

The foundation of the 2026 regulations rests upon a radically revised Power Unit (PU) architecture that fundamentally alters the energy economics of the vehicle. While the internal combustion engine (ICE) remains a 1.6-liter, 90-degree V6 with a single turbocharger, its functional role has been significantly redefined [**?**]. Most notably, the regulations mandate the complete removal of the Motor Generator Unit-Heat (MGU-H), eliminating a highly complex, expensive, and non-road-relevant component that previously provided near-continuous electrical harvesting from exhaust gases.

To compensate for the loss of the MGU-H and aggressively expand the role of electrical propulsion, the regulations dramatically uprate the remaining Motor Generator Unit-Kinetic (MGU-K). The maximum deployment power of the MGU-K is nearly tripled to 350 kilowatts (kW), establishing a near-perfect 50/50 power distribution between the internal combustion and electrical systems. To sustain this immense electrical output, the Energy Store (ES), or battery pack, has been conceptually scaled to manage approximately 4 megajoules (MJ) of deliverable energy per lap. This necessitates exceptionally high charge and discharge rates, presenting a formidable challenge in cell chemistry, thermal management, and high-voltage control systems.

Furthermore, the 2026 regulations enforce a strict environmental mandate concerning the chemical energy source of the ICE. The championship requires the exclusive use of 100% fully sustainable, advanced "drop-in" synthetic fuels [**?**]. This requirement fundamentally alters combustion characteristics, fuel mass flow limitations, and thermodynamic efficiency tar-

gets, compelling powertrain engineers to optimize the ICE specifically for synthetic hydrocarbon structures while maintaining the rigorous durability requirements of the sporting regulations.

2.2 Aerodynamic Changes

Complementing the power unit overhaul is a completely reimagined aerodynamic framework designed to mitigate the inherent challenges of the new powertrain while improving the racing spectacle. The primary objective of the 2026 aerodynamic regulations is to significantly reduce overall vehicle drag to compensate for the altered power delivery profile—specifically, the potential for early electrical energy depletion and subsequent velocity drop-offs on long straights.

To achieve this, the FIA has mandated a simplification of the external bodywork [**?**]. The structural dimensions of the vehicle are contracted, featuring a shorter wheelbase and a narrower overall width. These dimensional reductions inherently decrease the frontal area of the vehicle.

More critically, the regulations introduce fully integrated active aerodynamics as a cornerstone of the chassis design. The vehicles will employ movable front and rear wing elements that operate in distinct configurations: a high-downforce state for cornering and a low-drag state for straight-line velocity [**?**]. This active system is essential for achieving the aggressive regulatory targets of reduced downforce and significantly lowered drag compared to the previous vehicle generation. Despite these reductions, the fundamental aerodynamic philosophy continues to emphasize the underbody and ground effect phenomena for downforce generation, albeit with tighter dimensional constraints on the floor and diffuser to control the wake structure and maximize the aerodynamic efficiency of the vehicle trailing behind.

2.3 Weight and Packaging Constraints

The interplay between the expanded electrical powertrain and the contracted chassis dimensions creates an exceptionally constrained packaging environment. The 2026 regulations introduce stringent mass and volumetric limitations to ensure safety, control costs, and dictate conceptual direction.

Despite the significant increase in electrical component mass—stemming from the uprated MGU-K, heavier phase cabling, and the chemically dense 4 MJ battery pack—the FIA has established a minimum weight target of 800 kg [**?**]. Achieving this mass limit represents one of the most profound engineering hurdles of the regulatory cycle, demanding rigorous weight optimization across all structural and mechanical assemblies to prevent performance-inhibiting mass penalties.

A critical factor contributing to overall vehicle weight management is the substantial reduction in the permissible race fuel load. Facilitated by the 50/50 power split and the inherently higher efficiency of the long electrical deployment phases, the maximum fuel mass allowed for a standard race distance is reduced precipitously from 110 kg to just 70 kg [**?**]. This reduction allows for a significantly smaller fuel cell, structurally altering the monocoque topology and center of gravity behind the driver.

The packaging constraints are further exacerbated by uncompromising safety mandates. The massive 4 MJ Energy Store, along with the high-voltage control electronics, must be compactly packaged within a strictly defined safety volume inside the survival cell, primarily located beneath and behind the driver [**?**]. The spatial integration of these components, alongside the necessity for robust, high-capacity cooling networks to manage the intense thermal rejection of the 350 kW electrical architecture, defines the core packaging challenge of the 2026 Formula 1 car. This requires an

unprecedented level of spatial synergy between powertrain integration and structural chassis engineering.

3 Vehicle Architecture Overview

The fundamental architecture of this 2026-compliant Formula 1 concept is dictated by the delicate interplay between the radically revised, highly electrified power unit and the restrictive aerodynamic and packaging envelopes mandated by the FIA. The vehicle employs a traditional rear-mid engine layout, optimized specifically to accommodate the structurally significant 4 megajoule (MJ) Energy Store while adhering to the stringent 800 kg minimum weight and aggressively shortened dimensional targets. This section details the macroscopic packaging strategy, mass distribution, and primary structural subsystems that define the vehicle's dynamic baseline.

Table 1: Vehicle Architecture Design Parameters

Parameter	**Value**
Wheelbase	3580 mm
Overall Length	5125 mm
Vehicle Width	2000 mm
Front Track	1600 mm
Rear Track	1620 mm
CoG Height	270 mm
Weight Distribution	45% F / 55% R
Ground Clearance (F/R)	32 mm / 30 mm
Driver Recline	42°
Gearbox Length	590 mm
Total Mass	800 kg
Fuel Load	70 kg
Battery Energy	∼4 MJ

Figure 1: Full Vehicle Architecture and Packaging Diagram (Placeholder).

3.1 Dimensional Constraints and Mass Properties

Responding to the 2026 regulations aimed at enhancing agility and reducing overall vehicle footprint, the wheelbase is constrained exactly to 3580 mm, dictating a highly compact internal arrangement. The overall vehicle length spans 5125 mm, bounded by the strict aerodynamic dimensional boxes. Achieving the 800 kg minimum weight target represents a profound engineering challenge given the massive inclusion of the uprated 350 kW Motor Generator Unit-Kinetic (MGU-K) and the dense 4 MJ battery.

To maximize dynamic response, the vehicle achieves a static weight distribution of 45% front and 55% rear. This rearward bias naturally complements the rear-wheel propulsive characteristics of the dominant hybrid architecture while providing sufficient front-end authority for initial turn-in phases. A crucial performance differentiator is the Center of Gravity (CoG), which has been aggressively driven down to a height of 270 mm. This exceptionally low CoG minimizes lateral and longitudinal load transfer during dynamic maneuvers, directly enhancing mechanical grip. To facilitate this low CoG and further minimize the front-facing aerodynamic profile, the driver is positioned with an extreme 42-degree recline angle. Furthermore, the baseline static ride heights are set at 32 mm at the front axle and 30 mm at the rear, establishing a highly sensitive aerodynamic platform heavily reliant on stiff suspension architectures.

3.2 Powertrain and Energy Packaging

The core of the vehicle's motive force is the rear-mid mounted hybrid powertrain. The thermal component remains a 1.6-liter, 90-degree V6 internal combustion engine (ICE) featuring a single turbocharger, burning 100% sustainable synthetic fuel within a reduced 70 kg volumetric fuel cell. This cell is structurally integrated into the monocoque immediately behind the driver's bulkhead.

The defining characteristic of the 2026 architecture is the radical electrification of the propulsive system. The 350 kW MGU-K is packaged tightly alongside the ICE block, requiring significant localized structural reinforcements. The pivotal packaging challenge—the massive ∼4 MJ Energy Store—is integrated at the lowest extremity of the chassis floor beneath the fuel cell. Housing this heavy, dense component as low and centrally as possible is the primary enabler of the 270 mm CoG target. Power from the internal combustion and electrical systems is consolidated through a longitudinally mounted, 590 mm semi-automatic, seamless-shift gearbox casting.

3.3 Aerodynamic Architecture and Thermal Management

Conforming to the 2026 visual and functional mandates, the external aerodynamic surfaces are fundamentally simplified. The primary downforce generation mechanism relies acutely on the underbody floor structures, exploiting ground effect phenomena within the stringent volumetric boxes permitted downstream of the drastically simplified front wing assemblies.

Above the floor, the sidepods feature a minimalist, heavy-undercut profile. This geometry serves a dual purpose: aggressively channeling high-energy, undisturbed airflow down aggressively contoured flanks toward the critical floor edges and diffuser expansion zones, while simultaneously managing the intense thermal rejection requirements of the power unit. The undercut inlets feed structurally dense, highly efficient primary heat exchangers. Crucially, the extreme thermal loads generated by the MGU-K and the rapid charge/discharge cy-

cles of the larger Energy Store necessitate a discrete, secondary dielectric thermal management loop operating independently from the primary ICE cooling architecture.

Finally, the electronic architecture is highly centralized to minimize long, heavy, high-voltage cable runs. The Engine Control Unit (ECU) is mounted within the protective survival cell adjacent to the driver, while the heavy, high-current phase power electronics are integrated immediately adjacent to the MGU-K housing, ensuring minimal inductive losses between the battery, inverter, and motor systems.

4 Aerodynamics

The aerodynamic philosophy defining the 2026 Formula 1 technical cycle represents a paradigm shift necessitated by the aggressive electrification of the power unit. The aerodynamic framework, delineated by the FIA Technical Regulations [**?**], mandates a holistic reduction in overall vehicle drag to offset the depletion characteristics of the 4 megajoule (MJ) Energy Store and the heavily reliant 350 kW Motor Generator Unit-Kinetic (MGU-K). Concurrently, the regulations stipulate a reduction in absolute aerodynamic downforce to improve the "wake" profile of the leading car, facilitating closer circuit racing. This section details the macroscopic aerodynamic structures, the dominant ground effect mechanisms, and the critical integration of dynamic, active aerodynamic surfaces.

4.1 Front Wing

The 2026 regulations mandate a radically simplified multi-element front wing structure [**?**]. Compared to the preceding generation, the permissible number of aerodynamic elements is drastically reduced, severely restricting the wing's capacity for complex flow conditioning and localized high-pressure generation.

A primary focus in the design of this simplified structure is the optimization of the endplate geometry. Due to the narrowed 1600 mm front track width, the front wing must rapidly condition the freestream airflow directly ahead of the front tire face. The design necessitates a calculated trade-off between generating "outwash"—forcing turbulent tire wakes outward and away from the critical underfloor intakes—and centralized "inwash" to feed the core cooling structures. Given the stringent constraints on endplate outwash generation intended to improve the trailing car's aerodynamic environment [**?**], the wing profiles are heavily optimized to maintain robust attachment along the narrower span while managing the destructive tire squirt phenomenon.

4.2 Floor & Venturi Tunnels

With the aggressive simplification of upper aerodynamic surfaces, ground effect mechanics—specifically the underbody floor and venturi tunnels—re-emerge as the absolute primary source of total vehicle downforce [**?**]. The conceptual objective is to accelerate airflow through constricted underfloor channels, dropping static pressure and creating massive localized suction beneath the chassis.

The geometry of the venturi tunnels and the subsequent expansion rate of the rear diffuser angle are paramount. Due to the incredibly low 32 mm front and 30 mm rear dynamic ride heights, the floor is exceptionally sensitive to pitch and roll deviations. To maximize the extraction velocity through the diffuser, the leading edge of the floor requires the cleanest, highest-energy airflow available. This fundamental requirement dictates the geometry of the upstream bodywork; any turbulent, low-energy flow ingested by the venturi tunnels drastically degrades the suction peak and overall downforce coefficient (C_l).

Figure 2: Computational Fluid Dynamics (CFD) Flow Visualization of 2026 Aerodynamic Structures (Placeholder).

Table 2: 2026 Aerodynamic Targets and Operating Parameters

Parameter	**Target Specification**
Downforce Coefficient (C_l)	3.0 – 3.4
Drag Coefficient (C_d)	0.9 – 1.1
C_l/C_d Ratio (Aerodynamic Efficiency)	∼3.2
Front/Rear Downforce Split	35% F / 65% R
X-mode Drag Reduction (Active Aero)	∼15–20%
Static Ride Height (Front)	32 mm
Static Ride Height (Rear)	30 mm

4.3 Sidepods

To satisfy the fundamental requirement of delivering uncorrupted airflow to the floor edges and venturi inlets, the chassis utilizes a minimalist, heavy-undercut sidepod philosophy. The 2026 bodywork simplification rules heavily restrict the complex bargeboard and turning vane structures that historically conditioned this flow [**?**]. Consequently, the physical volume and shape of the sidepods themselves become the primary longitudinal flow conditioners.

By aggressively reducing the physical volume of the sidepods and employing steep undercut profiles, the design minimizes localized form drag. More critically, the undercut acts as a dedicated aerodynamic trench, accelerating high-pressure freestream air directly towards the upper surface of the floor and the critical diffuser expansion region. This macroscopic shape must be carefully balanced against the internal volumetric requirements for the primary heat exchangers, ensuring the cooling inlet sizing remains sufficient to manage the immense thermal rejection of the ICE while preserving the external aerodynamic efficiency.

4.4 Rear Wing

A defining visual and functional alteration in the 2026 aerodynamic regulations is the heavily simplified rear wing and the complete prohibition of the secondary "beam wing" structure [**?**]. The absence of the beam wing breaks the historical aerody-

namic link between the upper rear wing elements and the exiting flow of the floor diffuser.

Consequently, the remaining upper rear wing elements must be designed to operate highly independently. Without the beam wing to forcibly connect the low-pressure zone behind the rear wing with the diffuser exhaust, the overall rear downforce capacity is substantially reduced. The rear wing profiles are therefore optimized for maximal localized efficiency, carefully managing flow separation off the trailing edges to minimize the turbulent wake profile injected into the path of a following vehicle.

4.5 Active Aerodynamics

Perhaps the most significant technical innovation of the 2026 regulatory cycle is the mandated integration of fully Active Aerodynamics across both the front and rear wing assemblies [**?**]. This active system is not merely an overtaking aid (like the historical Drag Reduction System), but a fundamental requirement for the vehicle's standard operational lap profile.

The system operates strictly within two FIA-mandated states: the high-downforce "Z-mode" utilized through cornering phases, and the ultra-low-drag "X-mode" deployed on straightaways [**?**]. In Z-mode, the wing elements are closed, generating maximum C_l for mechanical grip. Upon corner exit, the system actuates into X-mode, significantly flattening the wing profiles and radically reducing the localized C_d.

This active modulation is inextricably linked to the 350 kW electrical deployment strategy. As the vehicle accelerates onto a straight, the MGU-K deploys its massive electrical torque. However, to prevent premature depletion of the 4 MJ Energy Store and devastating velocity clipping at the end of the straight, X-mode is engaged to shed aerodynamic resistance. Thus, the active aero effectively compensates for the altered power delivery profile, maintaining high terminal velocities while rationing electrical energy. The FIA mandates strict actuation constraints, defining specific track zones and dynamic thresholds where modes can be swapped to ensure safety and prevent uncontrolled aerodynamic instability.

4.6 Drag Reduction & Energy Interaction

Ultimately, the primary metric of success for a 2026 Formula 1 aerodynamic package is its interaction with the energy budget per lap. The macroscopic efficiency of the chassis—defined by the C_l/C_d ratio—directly dictates the duration and magnitude of electrical deployment.

The targeted 15–20% drag reduction achieved via X-mode actuation is critical. Lower inherent drag at high velocities directly reduces the resistive load on the powertrain. Consequently, the MGU-K does not need to deploy its maximum 350 kW output continuously simply to overcome aerodynamic resistance. By shedding drag mechanically, the vehicle requires less electrical energy to maintain its target velocity envelope. This extends the effective deployment range of the 4 MJ battery, mitigating severe derating events and directly yielding lower overall lap times. The aerodynamic design is therefore no longer evaluated solely on absolute downforce generation, but on its holistic symbiotic efficiency with the highly constrained electrical energy budget.

5 Hybrid Electric Power Unit Architecture

The paramount differentiator of the 2026 Formula 1 technical regulations lies in the total restructuring of the Power Unit (PU) [**?**]. The system transitions from an ICE-dominant architecture to a balanced, 50/50 split between internal combustion and electrical propulsion. The core philosophy demands massive electrical deployment capacity through an uprated Motor Generator Unit-Kinetic (MGU-K) operating within an approximate 4 megajoule (MJ) energy envelope, paired with a heavily constrained, sustainably fueled Internal Combustion Engine (ICE). This section examines the thermodynamic, structural, and control implications of the revised ICE underpinning this hybrid architecture.

5.1 Internal Combustion Engine

Despite retaining the foundational geometry of a 1.6-liter, 90-degree V6 with a single, centrally mounted turbocharger, the 2026 ICE is fundamentally distinct from its predecessors. The engine operates under a strict ∼15,000 RPM rev limit, with peak ICE output specifically legislated to fall within a restricted window of 400 to 450 kilowatts (kW) [**?**]. This reduction in absolute ICE power—down from the ∼630 kW peaks of the previous generation—is mathematically offset by the tripling of the MGU-K output to 350 kW, achieving the mandated 50/50 thermodynamic split.

The most transformative regulatory mandate regarding the ICE is the compulsory transition to 100% fully sustainable, "drop-in" synthetic fuels [**?**]. Synthesized from captured carbon and green hydrogen rather than extracted fossil deposits, these fuels possess distinctly altered molecular structures. Crucially, they exhibit significantly higher inherent oxygen content. This fundamental chemical shift forcibly alters the optimal stoichiometric air-to-fuel ratio (λ) required for complete combustion.

Table 3: 2026 Internal Combustion Engine Parameters

Parameter	**Value**
Configuration	1.6L V6 Turbo
Power Output	400–450 kW
Rev Limit	∼15,000 RPM
Fuel	100% Sustainable
MGU-H	Removed
PU Position	Rear-Mid
Gearbox Length	590 mm

Because the fuel is partially pre-oxygenated, the combustion event requires less atmospheric oxygen from the intake charge per unit mass of fuel. While this theoretically allows for higher fuel mass flow rates, the FIA strictly artificially limits total fuel energy flow to govern peak power [**?**]. Consequently, the combustion efficiency target requires a complete overhaul of engine mapping, ignition timing, and cylinder head design to extract maximum Brake Mean Effective Pressure (BMEP) from the synthetic hydrocarbon chains. The fundamental relationship governing the mechanical power output (P_{mech}) of the ICE remains:

$$P_{mech} = \frac{BMEP \cdot V_d \cdot N}{2} \tag{1}$$

where V_d is the displacement volume and

N is the engine speed. However, maximizing the overarching Brake Thermal Efficiency (η_{th}) is now heavily dependent on the lower heating value (LHV_{synth}) of the mandatory sustainable fuel:

$$\eta_{th} = \frac{P_{mech}}{\dot{m}_f \cdot LHV_{synth}} \tag{2}$$

where $\dot{m}_f$ represents the highly regulated fuel mass flow rate.

The complexity of optimizing this combustion process is brutally compounded by the regulatory removal of the Motor Generator Unit-Heat (MGU-H) [**?**]. In previous iterations, the MGU-H was structurally mounted to the turbocharger shaft. It served two critical functions: harvesting massive amounts of residual heat energy from the exhaust turbine to feed the battery, and acting as an electric motor to manually spool the compressor wheel, thereby entirely eliminating turbo lag during transient throttle applications.

The absence of the MGU-H fundamentally shifts the burden of low-RPM response back onto the ICE tuning and the MGU-K. Without the MGU-H to spin the compressor, the engine is entirely reliant on exhaust gas enthalpy to build boost pressure. To mitigate the severe anticipated turbo lag exiting slow corners, the 350 kW MGU-K must execute aggressive "torque fill" strategies—deploying massive electrical torque directly to the crankshaft to accelerate the vehicle while the turbocharger organically spools up to operating pressure. This demands unparalleled synchronization between the ICE fuel mapping and the algorithmic electrical deployment matrix.

Structurally, the ICE remains the foundational stressed member of the chassis. It utilizes a rear-mid positioning philosophy, bolting directly to the rear of the carbon fiber survival cell. The engine block and the structurally integrated 590 mm semi-automatic gearbox casing must tolerate the immense torsional loads transferred from the rear suspension elements. This compact, rearward consolidation of the heavy powertrain components—ICE, turbocharger, MGU-K, and gearbox—is meticulously engineered to maintain the critical 55% rear static weight distribution, ensuring maximum traction for the 350 kW electrical torque spikes, while the extreme low-packaging of the crankcase supports the aggressive 270 mm Center of Gravity (CoG) target required for dynamic agility.

5.2 MGU-K

The ascendancy of the Motor Generator Unit-Kinetic (MGU-K) is the most consequential performance metric of the 2026 regulations. The mandated peak output of 350 kW represents a massive, nearly 200% increase from the pre-2026 limitation of 120 kW [**?**]. This necessitates an exceptionally high-performance motor architecture; consequently, a Permanent Magnet Synchronous Motor (PMSM) is utilized due to its superior power density and efficiency at high rotor speeds compared to induction variants [**?**].

The operational torque (T_{mgu}) generated by the MGU-K is derived from the fundamental relationship:

$$T_{mgu} = \frac{P_{elec}}{\omega} \tag{3}$$

where P_{elec} is electrical power and ω is rotational velocity in rad/s. A critical analysis of this relationship reveals the MGU-K's torque contribution (T_{mgu}) is exponentially highest at low RPM. When the ICE is suffering from severe turbo lag at low engine speeds (e.g., exiting a hairpin), the MGU-K is capable of injecting crushing, instantaneous torque into the driveline to "fill the valley" left by the defunct MGU-H. As engine speed (ω) climbs, the available torque exponentially decays even at peak 350 kW deployment, handing the primary motive responsibility back to the spooling ICE.

Controlling this PMSM requires sophisticated field-oriented control (FOC) gov-

erned by the intrinsic motor characteristics. The back-electromotive force (EMF), denoted as E_{emf}, generated during high-speed rotation is given by:

$$E_{emf} = K_e \cdot \omega \tag{4}$$

where K_e is the back-EMF constant. The torque generated from the stator current (I_q) is determined by the torque constant (K_t):

$$\begin{aligned} T_m &= \frac{3}{2} p \lambda_m I_q \\ &= K_t \cdot I_q \end{aligned} \tag{5}$$

where p represents pole pairs and λ_m is the magnetic flux linkage.

Equally critical to deployment is the MGU-K's harvesting capacity under braking. During a typical heavy braking event (e.g., 5 distinct zones per lap), the MGU-K acts as a generator operating precisely at the 350 kW limit [**?**]. If the driver maintains this peak regenerative braking threshold for 2.5 seconds per zone, the raw energy harvested (E_{harv}) per zone is calculated as:

$$\begin{aligned} E_{harv} &= P_{regen} \cdot t_{brake} \\ &= 350,000\,\mathrm{W} \cdot 2.5\,\mathrm{s} \\ &= 875\,\mathrm{kJ} \end{aligned} \tag{6}$$

Extrapolated across 5 zones, this yields a raw harvest of 4.375 MJ—comfortably sustaining the 4 MJ Energy Store capacity. The deployment strategy is therefore highly cyclical: aggressively dumping energy on slow-corner exits to combat turbo lag, blending deployment through medium-speed curves to maintain traction matrices, and actively clipping energy delivery entirely on long straights via aero X-mode to preserve the battery cycle.

Figure 3: MGU-K Output Torque vs. ICE Torque Curve illustrating low-RPM Torque Fill Strategy (Placeholder).

5.3 Turbocharger System

The prohibition of the MGU-H mandates a return to a conventionally exhaust-driven, single turbocharger configuration [**?**]. In previous technical cycles, the MGU-H functioned as an electric motor coupled directly to the turbo shaft. Upon throttle application at low RPM (where exhaust gas volume is insufficient to spool the turbine), the MGU-H would draw electrical current to instantly spin the compressor, thereby entirely eliminating mechanical turbo lag. Without this assistance, the 2026 ICE suffers from a profound deficit in forced induction response.

This creates a brutal aerodynamic siz-

Table 4: 2026 MGU-K and Turbocharger Deployment Parameters

Parameter	**Value**
MGU-K Peak Power	350 kW
Pre-2026 MGU-K Power	120 kW
Motor Type	PMSM
Braking Zones per Lap	5
Harvest per Zone	875 kJ
Total Raw Harvest	4.375 MJ
Turbo Configuration	Single, No MGU-H
Lag Compensation	MGU-K Torque Fill

ing concession regarding the turbocharger itself. Designing a large turbocharger with a high-capacity compressor wheel facilitates greater peak power output (approaching the 450 kW ceiling) by flowing denser air masses at high engine speeds. However, the significantly higher rotational inertia of a large rotating assembly exacerbates the turbo lag issue exponentially. Conversely, a smaller turbo spooling rapidly to provide excellent drivability and low-speed response introduces a choking effect at high RPM, severely restricting the absolute power ceiling. This necessitates a compromised compressor map that prioritizes mid-range efficiency, relying entirely on the aforementioned MGU-K low-RPM "torque fill" to compensate for the delayed boost threshold.

The overall efficiency of the turbocharger assembly (η_{tc}) governs this delicate equilibrium, combining the isentropic efficiencies of both the compressor (η_c) and the turbine (η_t), minus mechanical bearing losses (η_m):

$$\eta_{tc} = \eta_c \cdot \eta_t \cdot \eta_m \tag{7}$$

Furthermore, the operational pressure ratio (Π_c) defined by the intake manifold boost pressure against ambient preconditions strictly governs the compressor mapping:

$$\Pi_c = \frac{P_{boost} + P_{atm}}{P_{atm}} \tag{8}$$

To manage this pressure ratio without MGU-H intervention, sophisticated wastegate strategies must control turbine speed to prevent damaging over-boost scenarios while attempting to maintain optimal momentum in the rotating group off-throttle.

6 Energy Storage System

The transition from a combustion-dominant formula to a true 50/50 power division in 2026 places an unprecedented burden on the electrochemical Energy Storage System (ESS) [**?**]. The battery pack is no longer a secondary boost mechanism but a primary propulsive unit, simultaneously required to manage near-instantaneous 350 kilowatt (kW) charge and discharge events within a volumetrically compact, 4 megajoule (MJ) energy envelope. This section derives the fundamental pack architecture, current demands, thermal loads, and the overarching dynamic lap energy balance from electrochemical first principles.

6.1 Cell Chemistry Selection

The selection of the underlying cell chemistry dictates the macro-level packaging and thermal viability of the entire powertrain. The required performance matrix prioritizes

extreme power density and high-rate cycling stability over absolute volumetric energy density.

Solid-state battery architectures were fundamentally rejected due to their current thermal immaturity; extracting 350 kW in transient pulses severely compromises the solid electrolyte interfaces at the mandated operational temperatures. Conversely, Lithium Iron Phosphate (LFP) chemistries, while highly stable, lack the requisite power density (W/kg) to satisfy the 350 kW draw without violating the 800 kg minimum chassis mass limitations.

Therefore, a high-power Nickel Manganese Cobalt (NMC) chemistry is selected [**?**]. NMC provides the superior specific power necessary for extreme transient discharge. The cells are packaged in a cylindrical form factor rather than flat pouches; the rigid structural casing of cylindrical cells better mitigates the severe mechanical expansion and contraction experienced under immense charge/discharge (C-rate) thermal cycling. The fundamental parameters of the selected NMC cell are a nominal voltage (V_{cell}) of 3.6 V (operating between 4.2 V max and 3.0 V min), a highly power-dense capacity (Q_{cell}) of 1.5 Ah, and an exceptionally low internal resistance (R_{cell}) of 5 mΩ.

6.2 Pack Architecture: 180S2P

The macro-level pack configuration is derived directly from the power unit inverter voltage targets and the FIA's 4 MJ storage limitation [**?**]. To minimize resistive (I^2R) losses across the high-voltage (HV) cabling while respecting modern inverter semiconductor breakdown thresholds, the nominal target pack voltage (V_{pack}) is established at 650 V.

The required number of cells connected in series (N_s) is calculated as:

$$\begin{aligned} N_s &= \frac{V_{pack}}{V_{cell}} \\ &= \frac{650\,\mathrm{V}}{3.6\,\mathrm{V}} \\ &\approx 180\,\mathrm{S} \end{aligned} \tag{9}$$

This yields an exact nominal pack voltage of:

$$\begin{aligned} V_{pack(actual)} &= 180 \cdot 3.6\,\mathrm{V} \\ &= 648\,\mathrm{V} \end{aligned} \tag{10}$$

The FIA mandates an operational deployable energy capacity (E_{pack}) of approximately 4 MJ. Converting this to standard electrochemical units:

$$\begin{aligned} E_{pack} &= \frac{4,000,000\,\mathrm{J}}{3600\,\mathrm{s}} \\ &= 1,111\,\mathrm{Wh} \approx 1.11\,\mathrm{kWh} \end{aligned} \tag{11}$$

With the required total energy (E_{pack}) and established pack voltage (V_{pack}), the required overall pack capacity (Q_{pack}) is:

$$\begin{aligned} Q_{pack} &= \frac{E_{pack}}{V_{pack(actual)}} \\ &= \frac{1111\,\mathrm{Wh}}{648\,\mathrm{V}} \\ &\approx 1.714\,\mathrm{Ah} \end{aligned} \tag{12}$$

Given the individual cell capacity (Q_{cell}) of 1.5 Ah, the required number of parallel strings (N_p) is 2. Therefore, the final derived pack architecture is **180S2P**, consisting of a total of 360 highly specialized NMC cylindrical cells.

6.3 C-Rate Analysis

The most brutal engineering metric of the 2026 ESS is the operational current draw defined by the 350 kW peak capability of the MGU-K. The peak pack current (I_{pack}) during full acceleration is:

$$
\begin{aligned}
I_{pack} &= \frac{P_{peak}}{V_{pack(actual)}} \\
&= \frac{350,000\,\mathrm{W}}{648\,\mathrm{V}} \\
&\approx 540\,\mathrm{A}
\end{aligned} \tag{13}
$$

Because the pack utilizes a 2P parallel configuration, this current is halved across the strings, resulting in a per-cell current (I_{cell}) of:

$$
\begin{aligned}
I_{cell} &= \frac{I_{pack}}{N_p} \\
&= \frac{540\,\mathrm{A}}{2} \\
&= 270\,\mathrm{A}
\end{aligned} \tag{14}
$$

The severity of this draw is quantified by the C-rate (C), calculated as the discharge current relative to total capacity:

$$
\begin{aligned}
C &= \frac{I_{cell}}{Q_{cell}} \\
&= \frac{270\,\mathrm{A}}{1.5\,\mathrm{Ah}} \\
&= 180\,\mathrm{C}
\end{aligned} \tag{15}
$$

To contextualize this extreme metric: consumer electric vehicles typically operate between 1 C and 3 C, while prior generations of high-performance motorsport (like Formula E) pushed boundaries near 50 C to 80 C. Discharging a chemical battery at 180 C represents the absolute frontier of electrochemical engineering. Such violence threatens rapid electrolyte vaporization and extreme degradation of the Solid Electrolyte Interphase (SEI) layer, cementing thermal management as the critical limiting factor of the vehicle.

6.4 Regenerative Energy Harvesting

The replenishment of the 180S2P pack relies entirely on the MGU-K operating as an electromagnetic retarder under braking. Assuming a profile of 5 major braking zones per lap, with the MGU-K harvesting at the mandated 350 kW peak limit for an average duration (t) of 2.5 seconds per zone, the energy harvested per zone (E_{zone}) is:

$$
\begin{aligned}
E_{zone} &= P_{regen} \cdot t \\
&= 350\,\mathrm{kW} \cdot 2.5\,\mathrm{s} \\
&= 875\,\mathrm{kJ}
\end{aligned} \tag{16}
$$

The total raw harvest energy (E_{raw}) per lap is therefore the summation across the n zones:

$$
\begin{aligned}
E_{raw} &= \sum_{i=1}^{n=5} (E_{zone})_i \\
&= 5 \cdot 875\,\mathrm{kJ} \\
&= 4375\,\mathrm{kJ} \\
&= 4.375\,\mathrm{MJ}
\end{aligned} \tag{17}
$$

This mathematical derivation reveals a critical system dynamic: the raw harvesting potential (4.375 MJ) explicitly exceeds the FIA-mandated storage capacity (4 MJ). Consequently, the Battery Management System (BMS) must algorithmically modulate or curtail regenerative harvesting during late-lap braking zones to prevent catastrophic over-voltage and thermal runaway within the almost-saturated cells.

6.5 Efficiency Chain: MGU-K to Battery

However, the raw harvested kinetic energy does not perfectly translate to stored chemical energy due to systemic thermal and switching losses. The cascaded efficiency chain (η_{total}) of the regenerative system is the product of the MGU-K motor efficiency ($\eta_m = 95\%$), the silicon carbide (SiC) inverter efficiency ($\eta_i = 97\%$), and the inherent coulombic efficiency of the NMC cells ($\eta_b = 98\%$) [**?**]:

$$\begin{aligned}\eta_{total} &= \eta_m \cdot \eta_i \cdot \eta_b \\ &= 0.95 \cdot 0.97 \cdot 0.98 \\ &\approx 0.903\end{aligned} \tag{18}$$

Applying this 90.3% cascaded efficiency to the raw lap harvest yields the actual net stored energy (E_{stored}):

$$\begin{aligned}E_{stored} &= E_{raw} \cdot \eta_{total} \\ &= 4.375\,\text{MJ} \cdot 0.903 \\ &\approx 3.95\,\text{MJ}\end{aligned} \tag{19}$$

Figure 4: Cascaded Efficiency Chain Block Diagram from Kinetic Recovery to Chemical Storage (Placeholder).

6.6 I^2R Losses and Thermal Design Load

The primary byproduct of the colossal 180 C discharge rate (Eq. 15) is intense Joule heating (I^2R loss) generated by the internal resistance of the pack configuration. With an individual cell internal resistance (R_{cell}) of 5 mΩ, the total series resistance (R_{series}) is:

$$\begin{aligned}R_{series} &= N_s \cdot R_{cell} \\ &= 180 \cdot 0.005\,\Omega = 0.9\,\Omega\end{aligned} \tag{20}$$

The parallel strings halve this total resistance, establishing the overall dynamic pack resistance (R_{pack}):

$$\begin{aligned}R_{pack} &= \frac{R_{series}}{N_p} \\ &= \frac{0.9\,\Omega}{2} = 0.45\,\Omega\end{aligned} \tag{21}$$

Consequently, when the MGU-K demands the peak 540 A draw (Eq. 13), the instantaneous thermal power dissipated (P_{loss}) entirely as heat within the core of the battery is:

$$\begin{aligned}P_{loss} &= I_{pack}^2 \cdot R_{pack} \\ &= (540\,\text{A})^2 \cdot 0.45\,\Omega \\ &= 131,220\,\text{W} \approx 131\,\text{kW}\end{aligned} \tag{22}$$

This 131 kW metric represents the absolute minimum thermal management design load. It dictates the fundamental dimensions, flow rates, and dielectric fluid properties of the mandated, isolated battery cooling loop. If the sidepod primary heat exchangers fail to reject this 131 kW influx continuously, the 180S2P pack will rapidly exceed its maximum operational temperature envelope, triggering devastating software derates and instantaneous lap time degradation.

6.7 Lap Energy Budget

The culmination of these derivations forms the absolute net dynamic energy balance across a standardized racing lap. The vehicle recovers and stores a net total of 3.95

MJ (Eq. 19). However, if the powertrain algorithm dictates a 350 kW deployment for an average of 4 seconds exiting the 5 major traction zones, the required deployment demand (E_{demand}) is:

$$\begin{aligned} E_{demand} &= P_{deploy} \cdot t_{sector} \cdot n \\ &= 350\,\mathrm{kW} \cdot 4\,\mathrm{s} \cdot 5 \\ &= 7000\,\mathrm{kJ} = 7\,\mathrm{MJ} \end{aligned} \tag{23}$$

This creates a stark net electrical deficit (ΔE_{lap}):

$$\begin{aligned} \Delta E_{lap} &= E_{demand} - E_{stored} \\ &= 7\,\mathrm{MJ} - 3.95\,\mathrm{MJ} \\ &= 3.05\,\mathrm{MJ} \end{aligned} \tag{24}$$

Figure 5: Software-Mapped Lap Energy Deployment vs. Kinetic Harvesting Matrix (Placeholder).

To sustain continuous performance over a race distance, this 3.05 MJ deficit must be systematically supplemented dynamically by the ICE operating as a secondary generator during specific non-traction limited phases. The software mapping must delicately thread this needle: commanding aggressive depletion on low-speed corner exits for lap time, dialing back MGU-K integration through high-speed aerodynamics-dependent corners, and dynamically throttling regenerative braking to protect the 131 kW thermal limit all while adhering to the 4 MJ physical volume constraint.

7 Electrical Power Management

The realization of the 2026 50/50 power division mandates an electrical architecture capable of translating the massive 350 kilowatt (kW) potential of the Energy Storage System into precise, usable mechanical torque at the Motor Generator Unit-Kinetic (MGU-K) [**?**]. This section details the graduate-level power electronics, modulation theory, and state estimation algorithms required to safely and efficiently manage the 540 A Direct Current (DC) link within a high-frequency, severe thermal environment.

7.1 Power Electronics Architecture: Inverter Design

The inverter topology acts as the critical bridge between the 648 V DC pack and the 3-phase Permanent Magnet Synchronous

Table 5: Electrical Engineering Parameterization of the 2026 Energy Storage System

Parameter	**Value**
Pack Configuration	180S2P
Total Cells	360
Cell Chemistry	NMC High-Power
Nominal Cell Voltage	3.6 V
Pack Voltage	648 V
Cell Capacity	1.5 Ah
Cell Internal Resistance	5 mΩ
Pack Resistance	0.45 Ω
Usable Energy	∼4 MJ (1.11 kWh)
Peak Discharge Power	350 kW
Peak Pack Current	540 A
Cell Current (2P)	270 A
Cell C-Rate	**180 C**
Peak I^2R Loss	131 kW
MGU-K Efficiency	95%
Inverter Efficiency	97%
Coulombic Efficiency	98%
Total Chain Efficiency	90.3%
Raw Harvest per Lap	4.375 MJ
Net Stored per Lap	3.95 MJ

Motor (PMSM). Given the extreme 350 kW load, Silicon Carbide (SiC) MOSFETs are universally selected over legacy Insulated-Gate Bipolar Transistors (IGBTs) [**?**]. The SiC wide-bandgap architecture drastically minimizes switching losses at high pulse-width modulation (PWM) frequencies, offers superior high-temperature stability, and maintains lower $R_{ds(on)}$ conduction resistance, critical for surviving the 648 V DC link environment.

The selected topology is a standard 3-phase full-bridge inverter. With the peak DC current (I_{dc}) established at 540 A during maximum 350 kW deployment, the required phase Root Mean Square (RMS) current ($I_{phase(rms)}$) delivered to the MGU-K stators is derived as:

$$\begin{aligned} I_{phase(rms)} &= \frac{I_{dc}}{\sqrt{3}} \\ &= \frac{540\,\mathrm{A}}{\sqrt{3}} \\ &= 311.77\,\mathrm{A} \approx 312\,\mathrm{A_{rms}} \end{aligned} \tag{25}$$

where I_{dc} represents the peak draw from the battery pack.

The target switching frequency (f_{sw}) of the SiC MOSFETs is established at ∼20 kHz to balance acoustic noise, control loop bandwidth, and switching losses. The instantaneous switching power loss (P_{sw}) across the inverter bridge is approximated by:

$$P_{sw} = f_{sw} \cdot E_{sw} \cdot I_{peak} \tag{26}$$

where E_{sw} is the total switching energy (turn-on plus turn-off) per pulse, and I_{peak} is the instantaneous peak phase current. A fundamental engineering tradeoff exists:

while a higher f_{sw} allows for smoother sinusoidal current synthesis (minimizing motor torque ripple), it linearly scales P_{sw} to catastrophic levels when switching 540 A. Concurrently, the conduction losses (P_{cond}) generated during the "on" state of the SiC devices are:

$$P_{cond} = I^2_{phase(rms)} \cdot R_{ds(on)} \tag{27}$$

where $R_{ds(on)}$ is the drain-to-source on-resistance of the SiC die. The summation of P_{sw} and P_{cond} across all six switch positions constitutes the total inverter thermal load. This massive thermodynamic influx necessitates a dedicated, liquid-cooled cold plate specifically sized to extract the combined I^2R and switching heat flux without violating the ~175°C maximum junction temperature of the SiC chips.

7.2 DC Link Capacitor Sizing

A critically sized DC link capacitor (C_{dc}) is positioned immediately across the inverter's DC bus to absorb the violent current ripple generated by the high-frequency switching events, protecting the battery pack from high-frequency inductive voltage spikes. The required capacitance (C) is derived from the fundamental relationship governing voltage ripple (dV) over the switching period (dt):

$$\begin{aligned} C_{req} &= \frac{I_{dc_ripple} \cdot dt}{dV} \\ &= \frac{540\,\mathrm{A} \cdot (50 \cdot 10^{-6}\,\mathrm{s})}{10\,\mathrm{V}} \\ &= 0.0027\,\mathrm{F} = 2.7\,\mathrm{mF} \end{aligned} \tag{28}$$

where I_{dc_ripple} is approximated by the 540 A peak phase draw, dt is the inverse of the 20 kHz switching frequency ($\frac{1}{20{,}000\,\mathrm{Hz}}$ = 50 μs), and dV is the strictly allowable voltage fluctuation target of 10 V to prevent inverter undervoltage faults.

Applying an arduous motorsport safety factor of 2 to account for thermal degradation and extreme regenerative current transients, the final specified capacitance (C_{dc}) is 5.4 mF. Advanced film capacitor topology is unequivocally mandated over traditional electrolytic variants. Film capacitors provide drastically lower Equivalent Series Resistance (ESR) at 20 kHz, which is paramount for survival in the 131 kW thermal environment radiating from the battery chassis. The internal ESR power loss (P_{esr}) within the capacitor itself is:

$$P_{esr} = I^2_{dc_ripple(rms)} \cdot ESR \tag{29}$$

where $I_{dc_ripple(rms)}$ is the effective RMS ripple current. A high ESR would trigger rapid dielectric failure at 540 A peak draws.

7.3 Field Oriented Control (FOC)

Controlling the MGU-K at 15,000 RPM requires absolute decoupling of the torque-producing and flux-producing stator currents. This is achieved via Field Oriented Control (FOC) [**?**]. The algorithm begins by reading the instantaneous 3-phase currents (i_a, i_b, i_c) and reducing them using the Clarke Transformation to a 2-phase stationary orthogonal reference frame (α, β):

$$\begin{aligned} i_\alpha &= \frac{2}{3}\left(i_a - \frac{1}{2}i_b - \frac{1}{2}i_c\right) \\ i_\beta &= \frac{2}{3}\left(0 + \frac{\sqrt{3}}{2}i_b - \frac{\sqrt{3}}{2}i_c\right) \end{aligned} \tag{30}$$

where $i_a + i_b + i_c = 0$ in a balanced wye-connected system. Subsequently, the Park Transformation rotates the stationary $\alpha\beta$ vectors by the measured rotor electrical angle (θ_e) into a synchronous rotating reference frame (d, q):

Figure 6: Field Oriented Control (FOC) Block Diagram illustrating Clarke/Park Transforms, PI Regulators, and SVPWM Synthesis (Placeholder).

$$\begin{aligned} i_d &= i_\alpha \cos(\theta_e) + i_\beta \sin(\theta_e) \\ i_q &= -i_\alpha \sin(\theta_e) + i_\beta \cos(\theta_e) \end{aligned} \tag{31}$$

By aligning the d-axis perfectly with the permanent magnet flux linkage (λ_f), the complex AC motor equations collapse into distinct, easily controllable DC quantities. The decoupled voltage equations in the synchronous dq frame are:

$$\begin{aligned} V_d &= R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ V_q &= R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \lambda_f) \end{aligned} \tag{32}$$

where R_s is stator resistance, L_d and L_q are the direct and quadrature inductances, and ω_e is the electrical rotational velocity.

Crucially, the electromagnetic torque (T_e) is now governed by:

$$T_e = \frac{3}{2} p [\lambda_f i_q + (L_d - L_q) i_d i_q] \tag{33}$$

where p represents pole pairs. The FOC algorithm allows Proportional-Integral (PI) controllers to independently regulate V_d to control the flux field via i_d, and V_q to dictate outright torque via i_q.

7.4 Maximum Torque Per Ampere (MTPA)

For a surface-mounted, non-salient PMSM architecture commonly utilized for extreme power density, the direct and quadrature inductances are virtually identical ($L_d \approx L_q$). This fundamental characteristic drastically simplifies the Maximum Torque Per Ampere (MTPA) strategy [**?**]. Because the reluctance torque component $[(L_d - L_q) i_d i_q]$ collapses to zero, the optimal MTPA trajectory strictly enforces a d-axis current setpoint of zero:

$$i_d^* = 0 \tag{34}$$

Consequently, all available stator current is directed exclusively to the q-axis to maximize torque output. Substituting $i_d = 0$ into Eq. 33 yields the simplified maximum torque relationship:

$$T_{max} = \frac{3}{2} p \cdot \lambda_f \cdot I_{max} \tag{35}$$

where I_{max} represents the previously derived 312 A RMS limit limit.

However, as the MGU-K accelerates toward its 15,000 RPM threshold, the induced back-EMF ($E = K_e \cdot \omega$) rapidly approaches the 648 V DC link voltage limit of the inverter. Once ω_e exceeds the base speed

where back-EMF equals the terminal voltage, the motor enters the Field Weakening region. To continue accelerating, the FOC algorithm must actively drive i_d negative. This negative d-axis current generates an opposing magnetic field that conceptually "weakens" the fixed permanent magnet flux (λ_f), thereby suppressing the back-EMF at the cost of reduced maximum i_q capacity and lower torque. The requisite field weakening trajectory is defined continuously by observing the voltage limit ellipse.

7.5 Space Vector PWM (SVPWM)

Translating the PI controller V_d, V_q reference voltages back into physical gating commands for the SiC MOSFETs relies on Space Vector Pulse Width Modulation (SVPWM). SVPWM is overwhelmingly preferred in motorsport over standard Sinusoidal PWM (SPWM) because it provides roughly 15% higher utilization of the DC bus voltage, unlocking an absolute peak fundamental phase voltage of $V_{dc}/\sqrt{3}$ compared to $V_{dc}/2$ for SPWM [**?**].

Figure 7: Space Vector PWM (SVPWM) Voltage Vector Hexagon with Sector Identification (Placeholder).

The SVPWM algorithm synthesizes the target voltage reference vector (V_{ref}) by dynamically switching between eight fundamental states of the 3-phase bridge: six active non-zero voltage vectors (V_1 through V_6) forming a hexagon, and two zero vectors (V_0, V_7) located at the origin.

The algorithm first identifies which of the six 60° sectors the rotating V_{ref} currently occupies based on its angle θ relative to the α-axis. The precise duty cycle timings required for the two adjacent active vectors (T_1, T_2) to geometrically synthesize V_{ref} within the switching period (T_s) are derived as:

$$T_1 = T_s \cdot \frac{\sqrt{3} \cdot V_{ref}}{V_{dc}} \cdot \sin\left(\frac{\pi}{3} - \theta\right) \tag{36}$$

$$T_2 = T_s \cdot \frac{\sqrt{3} \cdot V_{ref}}{V_{dc}} \cdot \sin(\theta) \tag{37}$$

The remaining fraction of the switching period is populated by the zero vectors (T_0):

$$T_0 = T_s - (T_1 + T_2) \tag{38}$$

The switching sequence is then meticulously arranged symmetrically (e.g., $V_0 \rightarrow V_1 \rightarrow V_2 \rightarrow V_7 \rightarrow V_2 \rightarrow V_1 \rightarrow V_0$) to minimize the absolute number of physical SiC switching transitions per cycle, drastically mitigating the formidable P_{sw} thermal load defined in Eq. 26.

7.6 State of Charge Estimation

Safely extracting laps from the exact 4 MJ energy limit requires highly rigorous State of Charge (SOC) tracking. The baseline measurement utilizes Ah-integration, or Coulomb counting:

$$SOC(t) = SOC(0) - \frac{1}{Q_{nom}} \int I(t)dt \quad (39)$$

where $SOC(0)$ is the initial charge state, Q_{nom} is the nominal pack capacity (1.714 Ah), and $I(t)$ is the dynamic discharge current. However, sensor noise and extreme 180 C current transients induce profound integration drift over a 50-lap race distance.

To continuously correct this drift, an Extended Kalman Filter (EKF) is implemented symbiotically with a Thevenin Equivalent Circuit Model (ECM) of the battery [**?**]. The ECM estimates the terminal voltage ($V_{terminal}$) based on the Open Circuit Voltage (OCV) curve:

$$V_{terminal} = OCV(SOC) - I \cdot R_0 - V_{RC} \quad (40)$$

where R_0 is immediate ohmic resistance, and V_{RC} characterizes the voltage drop across parallel resistor-capacitor diffusion networks mapping polarization physics. The EKF observer compares this mathematically predicted $V_{terminal}$ against the physical sensor voltage measurement, actively updating the internal covariance matrices to correct the predicted $SOC(t)$ value in real-time, eliminating integration drift.

Crucially, to prevent rapid degradation of the NMC chemistry, the BMS firmware hard-codes a strict operational SOC window between 20% and 95%. Therefore, the true effective usable SOC spans only 75% of the physical 1.11 kWh (4 MJ) pack structure.

$$\begin{aligned} E_{usable} &= (0.95 - 0.20) \cdot E_{pack} \\ &= 0.75 \cdot 1.11\,\text{kWh} \\ &\approx 0.833\,\text{kWh} \approx 3.0\,\text{MJ} \end{aligned} \quad (41)$$

This severe 3.0 MJ usable constraint fundamentally redefines the lap deployment geometry.

7.7 Energy Deployment Strategy

Recalculating the macroscopic lap energy budget against the 3.0 MJ usable boundary reveals a drastic operational deficit. As derived previously, 5 zones of optimal 350 kW 4-second deployments yield a gross demand (E_{demand}) of 7.0 MJ per lap. While braking recovery nets 3.95 MJ, only 3.0 MJ can be safely utilized without violating the 20% floor degradation threshold.

Therefore, the internal combustion engine (ICE) must operate aggressively in load-levelling mode, generating a net power surplus to supplement the staggering 4.0 MJ deficit:

$$\begin{aligned} \Delta E_{lap(ICE)} &= E_{demand_{gross}} - E_{usable} \\ &= 7.0\,\text{MJ} - 3.0\,\text{MJ} \\ &= 4.0\,\text{MJ} \end{aligned} \quad (42)$$

To govern this flow, the Central ECU executes a continuous trigraph state machine: 1. **HARVEST Mode**: Triggered under heavy braking. Kinetic energy flows backwards through the SiC inverter, regulated by the BMS to safely deposit the 3.95 MJ yield up to the 95% ceiling. 2. **DEPLOY Mode**: Triggered on slow corner exits. The MGU-K acts as a 350 kW motor, aggressively discharging the 3.0 MJ usable budget to enforce the "torque fill" turbo-lag compensation strategy. 3. **CONSERVE Mode**: Triggered on long, high-speed straights. The MGU-K deployment

Figure 8: FIA-Mandated Lap Energy Deployment State Machine Diagram (Placeholder).

Table 6: 2026 Power Electronics, FOC, and SOC Architecture Parameters

Parameter	**Value**
Inverter Topology	3-Phase Full Bridge
Switch Technology	SiC MOSFET
DC Link Voltage	648 V
Peak DC Current	540 A
Phase RMS Current	312 A
Switching Frequency	∼20 kHz
DC Link Capacitance	5.4 mF
Capacitor Type	Film
Motor Control	FOC + SVPWM
MTPA Strategy	$i_d = 0$ (pre-field weakening)
SOC Estimation	Coulomb Counting + Kalman Filter
Operational SOC Window	20% – 95%
Usable Energy (SOC-constrained)	3.0 MJ
Deployment Modes	Harvest / Deploy / Conserve
Isolation Voltage	648 V
Overcurrent Threshold (1.2×)	648 A
Thermal Cutoff	¿60°C

is actively clipped to zero. The chassis switches mapping to X-mode (drag reduction), and the ICE takes full propulsive dominance while concurrently spinning the MGU-K as a light generator to directly combat the 4.0 MJ deficit.

7.8 Protection and Isolation Systems

Managing a 648 V architecture atop a violently vibrating carbon-fiber chassis requires unrelenting, fail-safe isolation per motorsport standards [**?**]. Continuous Insulation Monitoring Devices (IMDs) inject high-frequency test signals to monitor chassis ground isolation. The minimum allowable isolation resistance (R_{iso_min}) is derived to prevent lethal shock under worst-case leakage current ($I_{leakage_max}$ typically ¡ 1mA):

$$R_{iso_min} = \frac{V_{dc}}{I_{leakage_max}} \tag{43}$$

Upon cold-start activation, the 5.4 mF DC link capacitor mimics a dead short. A dedicated pre-charge circuit utilizes a high-power resistor ($R_{precharge}$) to bypass the

main HV contactors, limiting the devastating inrush current (I_{inrush}) to a safe 10A target:

$$R_{precharge} = \frac{V_{dc}}{I_{limit}} = \frac{648\,\mathrm{V}}{10\,\mathrm{A}} = 64.8\,\Omega \quad (44)$$

Finally, catastrophic multi-tiered overcurrent thresholds trigger hardware gate-drive cutoffs at 1.2× peak nominal loads ($1.2 \cdot 540\,\mathrm{A} = 648\,\mathrm{A}$) within nanoseconds to arrest IGBT/SiC desaturation. Concurrently, software-level thermal cutoffs enforce a strict derating curve if continuous I^2R cycling drives cell temperatures (T_{cell}) above the 60°C threshold:

$$P_{limit} = P_{max} \cdot [1 - k_{derate}(T_{cell} - 60°\mathrm{C})] \quad (45)$$

where k_{derate} dictates the aggressive slope of the power truncation.

8 Thermal Management

The transition to a 50/50 Internal Combustion Engine (ICE) and electrical power division under the 2026 regulations generates an unprecedented amalgamation of concentrated heat fluxes [**?**]. Reconciling total vehicle heat rejection approaching 500 kW within the severely constrained, minimalist sidepod aerodynamic volumes dictates a complex, multi-loop thermal architecture. This section derives the fundamental thermodynamic loads across the five primary heat-producing subsystems.

8.1 ICE Cooling

The fundamental thermal boundary condition of the ICE is defined by the mandated 70 kg sustainable fuel limit. Utilizing an average Lower Heating Value (LHV) of 42 MJ/kg for the 100% sustainable drop-in fuel, the total chemical energy allocation per race is:

$$\begin{aligned} E_{race} &= m_{fuel} \cdot LHV \\ &= 70\,\mathrm{kg} \cdot 42\,\mathrm{MJ/kg} \\ &= 2940\,\mathrm{MJ} \end{aligned} \quad (46)$$

where m_{fuel} is the permitted fuel mass and LHV is the specific energy. Assuming a standard race duration of 90 minutes and a mean lap time of 70 seconds, the race distance constitutes 77 laps. The total chemical energy expended per lap (E_{lap}) is:

$$\begin{aligned} E_{lap} &= \frac{E_{race}}{N_{laps}} \\ &= \frac{2940\,\mathrm{MJ}}{77} \\ &\approx 38\,\mathrm{MJ} \end{aligned} \quad (47)$$

where N_{laps} is the total number of laps. With the V6 ICE operating at an estimated peak thermal efficiency (η_{th}) of 45%, the remaining 55% of the combustion energy is rejected as wasted heat ($Q_{rejected}$):

$$\begin{aligned} Q_{rejected} &= E_{lap} \cdot (1 - \eta_{th}) \\ &= 38\,\mathrm{MJ} \cdot 0.55 \\ &= 20.9\,\mathrm{MJ} \end{aligned} \quad (48)$$

where η_{th} is the brake thermal efficiency. Converting this per-lap energy into a continuous heat rejection rate (P_{reject}) yields:

$$\begin{aligned} P_{reject} &= \frac{Q_{rejected}}{t_{lap}} \\ &= \frac{20.9\,\mathrm{MJ}}{70\,\mathrm{s}} \\ &\approx 300\,\mathrm{kW} \end{aligned} \quad (49)$$

where t_{lap} is the average lap duration in seconds. This 300 kW thermal burden is roughly split equally: 150 kW is expelled directly out the tailpipe via exhaust gases, leaving 150 kW to be absorbed by the primary water-glycol coolant loop and dissipated through the sidepod radiators.

8.2 Battery Thermal Management

The Energy Storage System (ESS) is subjected to violent 180 C discharge rates, resulting in extreme Joule heating. As previously derived, the peak I^2R power loss (P_{peak}) across the 180S2P pack is:

$$\begin{aligned} P_{peak} &= I_{pack}^2 \cdot R_{pack} \\ &= (540\,\mathrm{A})^2 \cdot 0.45\,\Omega \\ &= 131\,\mathrm{kW} \end{aligned} \tag{50}$$

where I_{pack} is the peak DC current and R_{pack} is the total dynamically operative pack resistance. However, this 131 kW load is heavily transient. A single 70-second lap features 5 deployment phases of 4 seconds (20s total) and 5 harvesting phases of 2.5 seconds (12.5s total). Therefore, the total active time (t_{active}) at the peak 540 A load is 32.5 seconds. For the remaining passive phase ($t_{passive} = 37.5\,\mathrm{s}$), a nominal baseline load of ∼50 A is assumed for ancillary systems, yielding a passive power loss ($P_{passive}$) of:

$$\begin{aligned} P_{passive} &= I_{base}^2 \cdot R_{pack} \\ &= (50\,\mathrm{A})^2 \cdot 0.45\,\Omega \\ &\approx 1.1\,\mathrm{kW} \end{aligned} \tag{51}$$

where I_{base} is the ancillary current draw. The time-averaged steady-state heat rejection requirement (Q_{avg}) for the battery loop is calculated as:

$$\begin{aligned} Q_{avg} &= \frac{P_{peak} \cdot t_{active} + P_{passive} \cdot t_{passive}}{t_{lap}} \\ &= \frac{(131\,\mathrm{kW} \cdot 32.5\,\mathrm{s}) + (1.1\,\mathrm{kW} \cdot 37.5\,\mathrm{s})}{70\,\mathrm{s}} \\ &\approx 61.4\,\mathrm{kW} \end{aligned} \tag{52}$$

This derivation dictates strict thermal design targets [**?**]: the ESS loop must possess a transient peak absorption capacity of 131 kW but only requires a steady-state radiator sizing of 61.4 kW. Because the NMC chemistry requires a precise 20-40°C operating window—far below the ICE coolant temperature—the ESS mandates an entirely independent, dedicated liquid cold-plate loop circulating specialized dielectric fluid to guarantee temperature uniformity across all 360 cells.

8.3 Inverter Thermal Management

The SiC inverter bridges the 350 kW electrical transmission. Assuming an inverter efficiency (η_i) of 97%, the total dissipated thermal loss (P_{inv_loss}) is:

$$\begin{aligned} P_{inv_loss} &= P_{output} \cdot (1 - \eta_i) \\ &= 350,000\,\mathrm{W} \cdot 0.03 \\ &= 10.5\,\mathrm{kW} \end{aligned} \tag{53}$$

where P_{output} is the peak MGU-K mechanical power output. This 10.5 kW load is concentrated almost entirely within the microscopic surface area of the SiC die junctions. Conduction losses (P_{cond}) for the 3-phase bridge (accounting for 2 switches per phase leg) are:

$$\begin{aligned} P_{cond} &= 3 \cdot I_{phase}^2 \cdot R_{ds(on)} \cdot 2 \\ &= 3 \cdot (312\,\mathrm{A})^2 \cdot 0.005\,\Omega \cdot 2 \\ &\approx 2.9\,\mathrm{kW} \end{aligned} \tag{54}$$

where I_{phase} is the RMS phase current and $R_{ds(on)}$ is the die on-resistance. The switching losses (P_{sw}) across all 6 devices are approximated as:

$$\begin{aligned} P_{sw} &= f_{sw} \cdot E_{sw} \cdot 6 \\ &= 20,000\,\mathrm{Hz} \cdot 0.002\,\mathrm{J} \cdot 6 \\ &= 240\,\mathrm{W} \end{aligned} \tag{55}$$

where f_{sw} is the switching frequency and E_{sw} is the switching energy. The remaining ∼7.4 kW comprises gate drive, anti-parallel

diode, and stray inductive losses. Dissipating 10.5 kW from such a minuscule area yields an extreme local heat flux, demanding a direct-contact liquid-cooling heatsink designed to maintain the SiC junction temperatures strictly below 125°C to guarantee reliability across a race distance.

8.4 MGU-K Motor Thermal Management

Operating at a 95% efficiency (η_m), the MGU-K motor sheds the remaining 5% of the 350 kW load as heat (P_{mgu_loss}):

$$\begin{aligned} P_{mgu_loss} &= P_{output} \cdot (1 - \eta_m) \\ &= 350,000\,\mathrm{W} \cdot 0.05 \\ &= 17.5\,\mathrm{kW} \end{aligned} \tag{56}$$

This continuous 17.5 kW burden is comprised of copper, iron, and mechanical losses. The stator copper I^2R losses (P_{copper}) are:

$$\begin{aligned} P_{copper} &= 3 \cdot I_{phase}^2 \cdot R_{phase} \\ &= 3 \cdot (312\,\mathrm{A})^2 \cdot 0.002\,\Omega \\ &\approx 585\,\mathrm{W} \end{aligned} \tag{57}$$

where R_{phase} is the winding resistance per phase. However, at 15,000 RPM, the electrical frequency (f_{elec}) for a typical 4-pole-pair machine reaches:

$$\begin{aligned} f_{elec} &= \left(\frac{N}{60}\right) \cdot p \\ &= \left(\frac{15,000}{60}\right) \cdot 4 \\ &= 1000\,\mathrm{Hz} \end{aligned} \tag{58}$$

where N is the mechanical RPM and p represents the pole pairs. At this extreme 1000 Hz operative frequency, hysteresis and eddy current iron losses fundamentally dominate the stator core (∼8 kW). Windage and friction at 15,000 RPM account for an additional ∼2 kW, with the remaining ∼7 kW relegated to harmonic and stray losses. To respect the 180°C Class H insulation limit, the MGU-K utilizes direct oil-spray cooling on the stator end-windings and a hollow shaft channel to extract heat from the permanent magnet rotor assembly.

8.5 Turbocharger & Intercooler

The removal of the MGU-H dictates a fundamental redesign of the forced induction thermal management. The unassisted compressor must violently pressurize the intake charge, vastly increasing its operating temperature. Using the isentropic relation for an ideal gas, the ideal compressor outlet temperature (T_{2_ideal}) is:

$$\begin{aligned} T_{2_ideal} &= T_1 \cdot \pi^{\frac{\gamma-1}{\gamma}} \\ &= 298\,\mathrm{K} \cdot (4.0)^{0.286} \\ &\approx 443\,\mathrm{K} \end{aligned} \tag{59}$$

where T_1 is the ambient inlet temperature (298 K ≈ 25°C), π is an assumed pressure ratio of 4.0, and γ is the specific heat ratio of air (1.4). Factoring in a realistic compressor efficiency (η_c) of 75%, the actual outlet temperature (T_{2_actual}) spikes to:

$$\begin{aligned} T_{2_actual} &= T_1 + \frac{T_{2_ideal} - T_1}{\eta_c} \\ &= 298\,\mathrm{K} + \frac{443\,\mathrm{K} - 298\,\mathrm{K}}{0.75} \\ &\approx 491\,\mathrm{K} \approx 218^\circ\mathrm{C} \end{aligned} \tag{60}$$

where η_c is the isentropic efficiency. To prevent devastating pre-ignition (engine knock), this 218°C charge air must be violently cooled back down to ∼45°C before entering the intake plenum. For a 1.6L V6 spinning at 15,000 RPM, the approximate air mass flow rate ($\dot{m}$) is:

$$\begin{aligned}\dot{m} &= V_d \cdot \left(\frac{N}{120}\right) \cdot \rho_{boost} \\ &= 1.6 \cdot 10^{-3}\,\mathrm{m}^3 \cdot \left(\frac{15000}{120}\right) \cdot 4\,\mathrm{kg/m}^3 \\ &\approx 0.8\,\mathrm{kg/s}\end{aligned} \tag{61}$$

where V_d is the engine displacement volume and ρ_{boost} is the compressed air density. The resulting intercooler heat rejection requirement (Q_{ic}) is therefore:

$$\begin{aligned}Q_{ic} &= \dot{m} \cdot C_p \cdot \Delta T \\ &= 0.8\,\mathrm{kg/s} \cdot 1005\,\mathrm{J/kg \cdot K} \cdot (218^\circ\mathrm{C} - 45^\circ\mathrm{C}) \\ &\approx 139\,\mathrm{kW}\end{aligned} \tag{62}$$

where C_p is the specific heat capacity of air at constant pressure, and ΔT is the required temperature drop across the core. This massive 139 kW load, operating at vastly different temperature gradients than the ICE block, mandates a dedicated water-air intercooler circuit. Meanwhile, the turbine side operates uncooled despite ∼950°C exhaust inlets, relying entirely on advanced ceramic thermal barrier coatings. The center bearing housing is oil-cooled, rejecting a minimal ∼5 kW load into the auxiliary oil system.

Figure 9: Integrated Thermal Architecture Diagram demonstrating four independent cooling loops and their respective sidepod core packaging (Placeholder).

Table 7: 2026 Power Unit and Drivetrain Thermal Management Heat Load Parameterization

Heat Source	**Peak Load**	**Average Load**	**Cooling Method**
ICE Coolant	150 kW	150 kW	Water-glycol, primary sidepod radiators
Battery I^2R	131 kW	61 kW	Liquid cold plates, dedicated dielectric l
Intercooler	139 kW	139 kW	Water-air, dedicated low-temp circuit
MGU-K Motor	17.5 kW	17.5 kW	Oil-spray winding control, hollow shaft c
Inverter SiC	10.5 kW	10.5 kW	Direct-contact liquid heatsink
Turbo Bearing	5 kW	5 kW	High-flow oil cooling
Total System Rejection	**∼453 kW**	**∼383 kW**	

8.6 Integrated Thermal Architecture

The culmination of these disparate thermal sources dictates a segmented, four-loop integrated cooling architecture. The system must distinctly isolate: (1) the primary 150 kW ICE coolant loop running at high temperatures, (2) the highly-sensitive 131 kW capable ESS dielectric cold-plate loop constrained to 40°C, (3) the 139 kW water-air charge cooling intercooler circuit, and (4) an auxiliary integrated loop servicing the 10.5 kW SiC inverter and 17.5 kW MGU-K motor via oil-to-water heat exchangers. Packaging the physical radiator cores for all four circuits within the heavily restricted 2026 sidepod undercut philosophy—all while preserving underfloor aerodynamic airflow—represents the paramount packaging challenge of the entire vehicle architecture.

9 Electronics & Control Systems

Governing the 50/50 Internal Combustion Engine (ICE) and electrical power split across varying dynamic track conditions requires an immensely robust, real-time computational architecture [**?**]. This systems-level section delineates the deterministic control networks, sensor topologies, and telemetry pipelines mandated to process roughly 500 MB of state-estimation data per lap.

9.1 Electronic Control Unit (ECU)

At the apex of the digital architecture sits the Electronic Control Unit (ECU). The FIA explicitly mandates the utilization of a standardized, homologated central ECU across the grid—specifically, the McLaren Applied MCA-11 unit [**?**]. This homologation eliminates software-based traction control loopholes while enforcing a standardized data-logging envelope for scrutineering.

The MCA-11 serves as the central brain of the vehicle, assuming sole responsibility for critical deterministic tasks: ICE volumetric mapping, transient fuel injection timing, ignition spark advancement, and the arbitration of instantaneous torque demands between the ICE and the 350 kW Motor Generator Unit-Kinetic (MGU-K).

To execute these tasks without catastrophic desynchronization, the ECU runs on a strictly deterministic Real-Time Operating System (RTOS). In this environment, there is zero latency tolerance on safety-critical channels; control loops must execute with microsecond precision. Consequently, all critical Input/Output (I/O) channels feature hardware-level dual redundancy. Furthermore, the ECU continuously interfaces with the mandatory FIA accident data logger, mirroring a tightly specified telemetry envelope for regulatory compliance checking.

9.2 Sensor Architecture

Feeding the ECU's state machine is a distributed architecture of approximately 300 discretely polled sensors. This array is rigorously categorized into four distinct measurement domains:

1. **Powertrain Sensors:** Measuring high-frequency ICE states including wideband lambda (exhaust oxygen) ratio, multipoint coolant and oil pressures/temperatures, exhaust gas temperatures (EGT) exceeding 950°C, turbocharger boost pressures, and direct MGU-K stator winding temperatures. 2. **Battery Management (BMS):** The Energy Storage System demands individual voltage (V_{cell}) and temperature (T_{cell}) monitoring across all 360 Nickel Manganese Cobalt (NMC) cells. It relies on a high-precision Hall-effect

pack current sensor and continuous chassis isolation monitoring. 3. **Chassis & Vehicle Dynamics:** Linear potentiometers at all four pushrod/pullrod corners track instantaneous suspension displacement. Triaxial accelerometers and gyroscopes monitor yaw, pitch, and roll rates. High-resolution brake line pressure transducers and infrared Tire Pressure and Temperature Monitoring Systems (TPMS) provide critical grip estimates. 4. **Navigation & Positioning:** A dual-band GPS antenna works symbiotically with a 6-axis Inertial Measurement Unit (IMU) to provide absolute track positioning and attitude.

Crucially, the raw data from the IMU and GPS are not utilized independently. A Sensor Data Fusion algorithm—typically a variant of an Extended Kalman Filter—mathematically merges the high-frequency inertial data with the low-frequency GPS coordinates to output an incredibly accurate, drift-free vehicle dynamics state estimation matrix.

9.3 Onboard Data Architecture

Transmitting this dense sensor payload requires a hardened digital backbone. Standard Controller Area Network (CAN) buses (capped at 1 Mbps) are insufficient for modern MGU-K control loops. Therefore, the 2026 architecture leverages a hybrid FlexRay and CAN FD (Flexible Data-Rate) topography [**?**]. CAN FD unlocks bandwidths up to 10 Mbps, while FlexRay guarantees the strict, time-triggered deterministic timing critical for high-voltage inverter switching.

The physical architecture is strictly segregated into isolated network domains: 1. **Powertrain Network:** A high-priority FlexRay bus linking the ECU, the Silicon Carbide (SiC) inverter, and the BMS to execute the 350 kW torque requests seamlessly. 2. **Chassis Network:** A CAN FD domain tracking suspension, active aerodynamic servos, and brake-by-wire hydraulic matrices. 3. **Safety Network:** A completely isolated, fault-tolerant loop governing fire suppression, Halo structural integrity sensors, and the FIA accident data recorder.

This physical segregation enforces a strict cybersecurity and safety envelope: a catastrophic electrical short in a suspension potentiometer on the Chassis network fundamentally cannot propagate to or interfere with the Safety or Powertrain domains. Combined, the onboard telemetry generates approximately 500 MB of logged data per lap, saturating the internal memory at a rate of roughly 500 Mbps.

9.4 Telemetry System

Simultaneously, a fraction of this heavily encrypted data is downlinked in real-time to the pitwall. To comply with FIA sporting regulations [**?**], the telemetry system utilizes a 2-4 Mbps bandwidth microwave downlink. This permits pitwall engineers and factory-based operations rooms to monitor approximately 1000 highly prioritized channels live.

The primary antennas are aerodynamically integrated directly into the carbon-fiber engine cover to preserve the laminar flow utilized by the rear wing. The uplink from the pitwall to the car is severely restricted; teams may only transmit permitted parameter adjustments (e.g., acknowledged pit-lane delta signals or standardized dashboard text). Two-way algorithmic adjustments during a race are strictly prohibited.

To prevent competitor interception of strategic variables, all transmitted data arrays are subjected to military-grade encryption payloads. Furthermore, the RF hardware guarantees an absolute latency requirement of < 100 ms, ensuring that pitwall feedback remains actionable when identifying thermal anomalies or deploying strategic undercuts.

9.5 Power Distribution Architecture

The vehicle electrical tapestry is divided transversally into two strictly isolated voltage domains, necessitated by galvanic safety standards.

The High-Voltage (HV) Traction Domain operates nominally at 648 V DC, encompassing the 180S2P battery pack, the MGU-K SiC inverter, and the step-down DC-DC buck converter.

The Low-Voltage (LV) Auxiliary Domain operates on standard 12 V and 24 V architectures, powering the ECU, the 300-sensor array, active aerodynamic hydraulic servos, and the telemetry transceivers.

Between these nets, a highly efficient, galvanically isolated DC-DC converter steps the 648 V traction battery down to 12/24 V, eliminating the need for a separate, heavy LV lead-acid battery. Standard mechanical fuses are entirely replaced by a solid-state Power Distribution Unit (PDU). The PDU utilizes programmable MOSFETs acting as solid-state relays, offering infinitely adjustable current tripping limits, microsecond cutoff times, and comprehensive fault logging.

Furthermore, engaging the HV domain requires a hardware pre-charge circuit [**?**]. Upon startup, the circuit routes the DC link through a high-power resistor to slowly charge the inverter's 5.4 mF film capacitor, limiting the inrush current (I_{inrush}) that would otherwise weld the main contactors:

$$I_{inrush} = \frac{V_{dc}}{R_{precharge}} \tag{63}$$

where V_{dc} is the 648 V pack voltage and $R_{precharge}$ is the 64.8 Ω resistor calculated in Section 8.

Figure 10: Electronics Architecture Block Diagram illustrating Network Domains, ECU, BMS, Inverter, Sensors, Telemetry, and Power Distribution Paths (Placeholder).

9.6 Software & Control Algorithms

The embedded firmware running on the MCA-11 follows a strictly layered, AUTOSAR-equivalent architecture to guarantee automotive-grade reliability. This prevents logical tangling between distinct functions.

1. **Hardware Abstraction Layer (HAL):** The lowest tier, directly interfacing with the physical world by running custom sensor drivers and interpreting analog signals or CAN FD packets. 2. **Real-Time Control Layer:** The deterministic execution block running the FOC motor control vectors and interpreting the ICE lambda and ignition maps to synthe-

Table 8: 2026 Vehicle Electronics, Sensors, and Telemetry Specification

Subsystem	**Key Specification**
Electronic Control Unit (ECU)	McLaren Applied MCA-11
Operating System Type	Deterministic Real-Time (RTOS)
Total Distributed Sensors	∼300
ESS Cell Monitors	360 (individual V_{cell} + T_{cell})
Internal Onboard Data Rate	∼500 Mbps
Logged Data per Lap	∼500 MB
Telemetry Downlink Bandwidth	2-4 Mbps (Encrypted)
Live Pitwall Telemetry Channels	∼1000
Telemetry RF Latency	<100 ms
High-Voltage (HV) Domain	648 V DC
Low-Voltage (LV) Auxiliary Domain	12 V / 24 V
Network Backbone Protocols	FlexRay + CAN FD
Segregated Network Domains	3 (Powertrain, Chassis, Safety)

size immediate torque. 3. **Energy Management Layer:** The strategic oversight block enforcing the SOC-based deployment strategy from Section 8, running the Harvest/Deploy/Conserve state machine derived from the 3.0 MJ usable constraint. 4. **Telemetry Layer:** An asynchronous background thread responsible for data packaging, military-grade algorithmic encryption, and microwave downlink scheduling.

All crucial calibrations—including granular ICE ignition timing maps, MGU-K transient torque curves, and brake-by-wire hydraulic balance maps—are flashed directly into the ECU's non-volatile memory, serving as the ultimate intellectual property of the team.

10 Chassis & Materials

Satisfying the paradox of the 2026 regulations—integrating a 350 kW electrical architecture and a 4 MJ Energy Storage System while aggressively adhering to an absolute 800 kg total vehicle mass target—demands extreme materials science [**?**]. This section details the structural engineering of the primary monocoque, the specific load-bearing alloys utilized, and the geometric packaging of the high-voltage hybrid components within the carbon structure.

10.1 Monocoque Design

The primary survival cell and structural backbone of the vehicle is the monocoque, constructed entirely from advanced Carbon Fiber Reinforced Polymer (CFRP). Extending continuously from the front suspension bulkhead to the primary firewall behind the driver, the tub is designed to act as an immensely rigid torsion box. Crucially, the V6 Internal Combustion Engine (ICE) acts as a fully stressed member of this chassis; the engine block bolts directly to the rear of the monocoque and directly carries the inboard suspension loads of the rear axle, completely eliminating the mass penalty of a traditional separate rear subframe.

Monocoque wall thickness varies drastically from 3 mm to 10 mm depending on the localized stress tensor of the specific zone. The flexural stiffness (EI) of these com-

posite panels is governed by the structural bending equation:

$$EI = E \cdot \frac{b \cdot t^3}{12} \tag{64}$$

where E is the Young's modulus of the composite laminate, b is the panel width, and t is the cross-sectional thickness. Because EI scales cubically with thickness, the ply stacking sequence is hyper-optimized per region.

In high-load, multi-directional stress zones—such as the front suspension pickup points—the layup utilizes a quasi-isotropic strategy (e.g., [0°/45°/90°/-45°] stacking) to ensure uniform load distribution regardless of the instantaneous suspension vector. Conversely, in the bending-dominant floor sills, heavily unidirectional [0°] plies are layered longitudinally to maximize the effective E in the primary load axis. This meticulous localized fiber alignment guarantees the stringent target torsional stiffness of $> 30,000$ Nm/deg, an absolute prerequisite for maintaining millimeter-perfect aerodynamic ride heights and suspension geometry consistency under 5G dynamic-aero loads.

10.2 Material Selection

Hitting the 800 kg limit entirely precludes the use of standard commercial-grade carbon fiber. The primary structural fiber selected across the monocoque is ultra-high modulus T800/T1000 grade polyacrylonitrile (PAN) based carbon fiber [**?**]. This aerospace-grade filament boasts a tensile modulus of 294 GPa and an ultimate tensile strength ranging from 5490 to 6370 MPa. The superior specific stiffness (E/ρ) and specific strength of T1000 carbon justify its immense cost, as it allows structural engineers to drastically reduce ply counts (and therefore mass) without sacrificing the rigidity derived in Eq. **??**.

The fiber matrix utilizes a toughened epoxy prepreg resin system, cured inside high-pressure autoclaves subjected to 120°C and 6-7 bar of atmospheric pressure to aggressively eliminate microscopic voids and ensure absolute fiber compaction. For non-structural aerodynamic bodywork and the bulk of the underfloor, a Nomex aramid honeycomb core is sandwiched between ultra-thin composite skins. This provides immense geometric shear stiffness and buckling resistance at a negligible density penalty (48 kg/m^3).

Metallic alloy usage is strictly relegated to components demanding extreme isotropic yield strengths or ultra-high thermal resistance. The mandatory safety Halo, suspension uprights, and highly stressed primary fasteners are machined from Ti-6Al-4V (Grade 5) titanium alloy [**?**], characterized by a formidable tensile strength of 950 MPa at a density of 4430 kg/m^3 (nearly half that of high-tensile steel). Finally, the brake discs and pads utilize a highly specialized Carbon-Carbon (C/C) composite matrix, uniquely capable of maintaining a stable, aggressive coefficient of friction even as rotor core temperatures exceed 1000°C.

Table 9: 2026 Primary Chassis, Structural, and Component Material Properties

Material	**Application**	**Tensile Strength**	**Density**
T800/T1000 CFRP	Primary Monocoque, Stressed Bodywork	5490-6370 MPa	1800 kg/m^3
Toughened Epoxy	Composite Resin Matrix	80-90 MPa	1200 kg/m^3
Nomex Honeycomb	Floor Cores, Non-Structural Panels	—	48 kg/m^3
Ti-6Al-4V Alloy	Halo Structure, Uprights, Fasteners	950 MPa	4430 kg/m^3
Carbon-Carbon	High-Temperature Brake Discs	—	1750 kg/m^3

10.3 Impact Structures

To protect the driver from catastrophic ballistic energy, the monocoque is surrounded by sacrificial composite impact structures. The front nose cone features a mandatory geometry defined by the FIA [**?**], acting as a progressive crush zone designed to instantaneously disintegrate and safely shed frontal impact energy. The kinetic energy absorbed ($E_{absorbed}$) during a collision is mathematically expressed as:

$$E_{absorbed} = \sigma_{crush} \cdot A \cdot \delta \tag{65}$$

where σ_{crush} is the dynamic composite crushing stress, A is the instantaneous cross-sectional area of the nose, and δ is the longitudinal crush distance. A similar, smaller structure is appended directly behind the 590 mm transverse gearbox to mitigate rear impacts.

Side impact structures (SIPS) consist of heavily reinforced, specialized CFRP tubes and layered impact panels integrated directly into the monocoque flanks adjacent to the driver. All impact elements must survive brutal FIA mandatory static load homologation testing prior to track operation: 60 kN of sustained axial force on the frontal structures, and 25 kN of lateral force on the side panels. The engineering challenge lies in optimizing the crush geometry (A and σ_{crush}) for maximum energy absorption while simultaneously minimizing the structure's physical aerodynamic footprint.

10.4 Safety Halo

The Titanium Ti-6Al-4V Halo is a mandatory upper safety structure engineered to prevent large debris from striking the driver's helmet. Weighing approximately 7 kg, the structure must withstand a staggering 125 kN static peak homologation load applied vertically from above.

The Halo legs are mechanically integrated deep into the CFRP monocoque sides, allowing the massive 125 kN external load vectors to transfer directly down into the primary tub without buckling the cockpit surround. The interface between the titanium pylons and the carbon tub represents the highest localized structural stress concentration zone on the entire vehicle, requiring intense reinforcement with specialized multidirectional carbon fiber plies to prevent galvanic corrosion and interlaminar delamination under sheer load.

To verify the Ti-6Al-4V geometry, the primary cross-sectional tensile stress (σ) exerted on the Halo structure is derived as:

$$\sigma = \frac{F}{A_{cross}} = \frac{125,000\,\mathrm{N}}{A_{cross}} \tag{66}$$

where F is the applied FIA static load, and A_{cross} is the structural cross-sectional area. By enforcing a strict safety factor against the 880 MPa yield strength of the Ti-6Al-4V alloy, engineers mathematically back-calculate the absolute minimum required A_{cross} for the Halo pillars, aggressively shaving excess titanium mass to lower the vehicle center of gravity (CoG).

10.5 Hybrid Component Packaging

Integrating a 648V, 4MJ battery pack into the physical carbon chassis requires exquisite electrical and structural geometry. The primary battery tub is physically integrated into the lowest possible stratum of the chassis floor, nestled precisely between the driver's legs and the primary firewall. This strategic placement ensures the immense density of the 360 NMC cells forcefully anchors the global vehicle CoG exactly at the optimal 270 mm aerodynamic target. Furthermore, the localized dielectric liquid cold plates required to suppress the 131 kW I^2R thermal loads (as derived in Section 10) are directly co-cured into the carbon floor structure to completely eliminate the mass of secondary heat-exchanger casings.

Figure 11: Monocoque Cross-Section Diagram illustrating CFRP Ply Orientation Zones, Battery Tub Integration, 648V Cable Routing limits, and Ti-6Al-4V Halo Attachment Points (Placeholder).

Routing the extreme 648 V, 540 A phase cables through a conductive carbon fiber chassis poses a severe galvanic and safety threat. All high-voltage internal cables are heavily shielded and externally coded perfectly orange per strict FIA and ICE electrochemical regulations. These heavy cables are routed through completely dedicated, completely isolated CFRP conduits laminated physically into the monocoque sidewalls.

To prevent catastrophic galvanic interaction and carbon tracking at 648 V—which would instantly electrify the entire structural tub—the geometry mandates that all exposed HV cable routing maintains a rigid clearance of > 50 mm from the raw structural carbon. Finally, the BMS logic boards are mounted immediately adjacent to the battery within a heavily shielded, temperature-conditioned zone to minimize signal degradation across the vital 360-channel balancing harness, protected from the ICE behind the mandatory flame-resistant composite physical firewall separating the driver's cockpit from the powertrain.